

# Emanuele Rossi

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Researcher working on machine learning for animal communication, part of a broader interest in leveraging ML to recover structure in complex systems: previously social graphs (Temporal Graph Networks, SIGN) at Twitter and Imperial College London, and generative models for biomolecules at VantAI.

**Google Scholar metrics:** 4,261 citations; h-index 15 (July 2026).

## Education

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**Imperial College London** Oct 2020 – Feb 2024

*Ph.D. in Computer Science – Supervisor: Prof. Michael Bronstein*

- Research focused on **graph machine learning methods** for large real-world systems

**University of Cambridge** Oct 2018 – Jun 2019

*Master in Advanced Computer Science, Distinction (4.0 GPA)*

**Imperial College London** Oct 2015 – Jun 2018

*Bachelor in Computer Science, First Class Honors (4.0 GPA)*

## Experience

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**Postdoctoral Researcher** Rome (remote)

*GLADIA Research Group* [🔗](#), Sapienza University of Rome – With Prof. E. Rodolà Oct 2025 – Present

- Developing machine learning methods for animal communication.

**Machine Learning Researcher** NYC (remote)

*VantAI* Jan 2024 – Aug 2025

- **Co-led development of Neo-1** [🔗](#), a state-of-the-art all-atom latent diffusion model for multi-modal structure prediction and de-novo generation of biomolecular complexes, trained on 100+ GPUs.

**Machine Learning Researcher** London

*Twitter* Jun 2019 – Feb 2023

- Developed **graph ML models** for Twitter's social graph and deployed a recommendation model that generated **1.2M additional daily follows**

**Machine Learning Researcher** London

*Fabula AI* Mar 2019 – Jun 2019

- Developed graph ML models for fake news detection as part of Fabula AI's small research team before [Twitter acquired the company](#) [🔗](#)

**Software Engineering Intern** California

*Google* Jul 2017 – Oct 2017

## Selected Publications

[Full list](#)

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### Animal Communication & Audio

- E. Rossi and E. Rodolà. [Communicating Sound Through Natural Language](#) [🔗](#). *arXiv* 2026.
- D. Marincione et al. [Model Merging Improves Zero-Shot Generalization in Bioacoustic Foundation Models](#) [🔗](#). *NeurIPS Workshop on AI for Animal Communication 2025, Spotlight (top 15%)*.

### Graph Learning

- E. Rossi et al. [Temporal Graph Networks for Deep Learning on Dynamic Graphs](#) [🔗](#). *ICML GRL Workshop 2020*. **1,700+ citations** and **1,000+ GitHub stars** [🔗](#).
- E. Rossi\*, F. Frasca\* et al. [SIGN: Scalable Inception Graph Neural Networks](#) [🔗](#). *ICML GRL Workshop 2020*. **600+ citations** and used in **production at Airbnb** [🔗](#).

- *B. Chamberlain, S. Shirobokov, E. Rossi et al.* [Graph Neural Networks for Link Prediction with Subgraph Sketching](#) [↗](#). *ICLR 2023, Oral (top 5%)*. **200+** citations.

## Leadership

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**Co-Founder, LeadTheFuture** ([Forbes article](#) [↗](#))

*Jun 2018 – Mar 2023*

- Built a non-profit mentoring community with more than **200 mentors** and **500 selected mentees**